KAIZEN MATHS

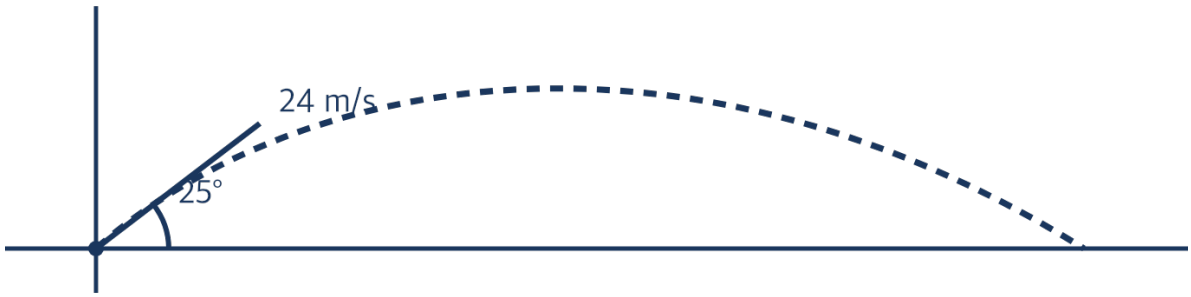
Lesson starter

Projectile Motion

Components & Basics

Name _____

SA1



A projectile is launched at 24 m/s at an angle of 25° above the horizontal. Find its horizontal and vertical components of velocity.

MEC-5B63-L1 · SA

Show your working

KAIZEN MATHS

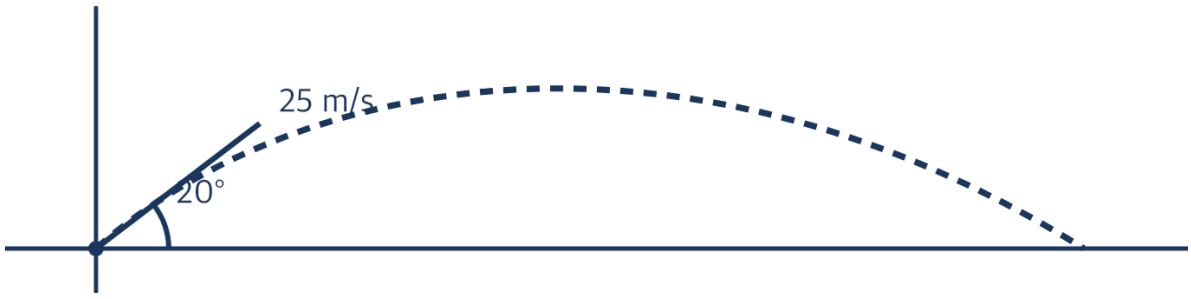
Lesson starter

Projectile Motion

Components & Basics

Name _____

SB1



A projectile is launched at 25 m/s at 20° above the horizontal. Find the time taken to reach maximum height.

MEC-5B63-L1 · SB

Show your working

KAIZEN MATHS

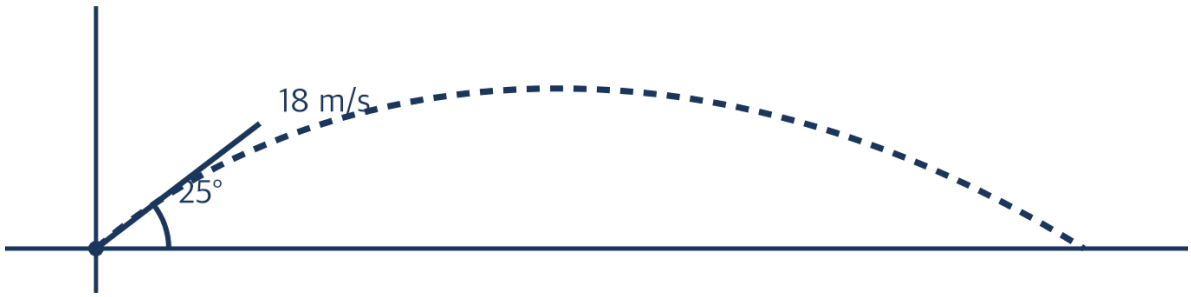
Lesson starter

Projectile Motion

Components & Basics

Name _____

SC1



A projectile is launched from level ground at 18 m/s at 25°. Find the total time of flight. Take g = 9.8 m/s².

MEC-5B63-L1 · SC

Show your working

KAIZEN MATHS

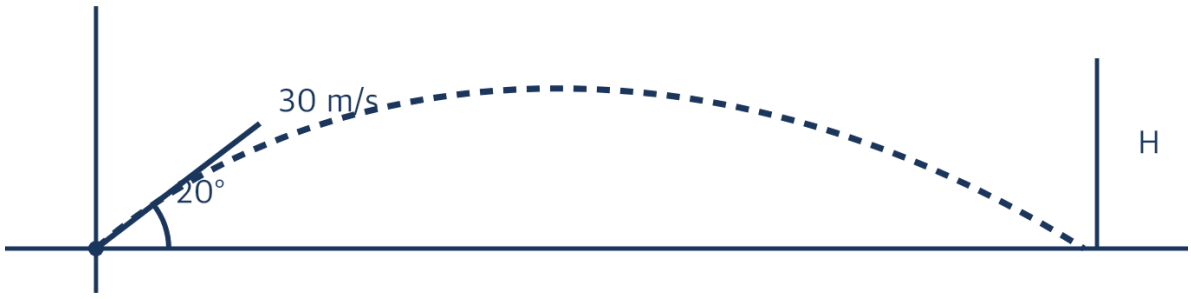
Lesson starter

Projectile Motion

Components & Basics

Name _____

SD1



A projectile is launched from level ground at 30 m/s at 20°. Find the maximum height reached.

MEC-5B63-L1 · SD

Show your working

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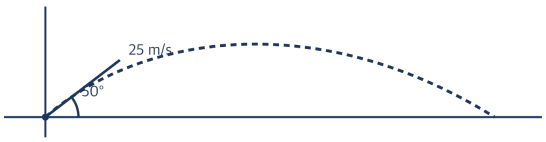
Exit ticket

Projectile Motion

Components & Basics

Name _____

EA1



A projectile is launched at 25 m/s at an angle of 50° above the horizontal. Find its horizontal and vertical components of velocity.

Show your working

KAIZEN MATHS

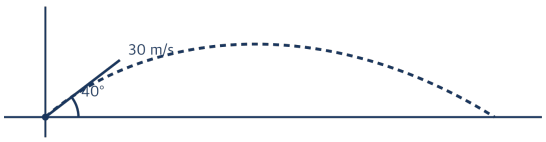
Exit ticket

Projectile Motion

Components & Basics

Name _____

EB1



A projectile is launched at 30 m/s at 40° above the horizontal. Find the time taken to reach maximum height.

Show your working

KAIZEN MATHS

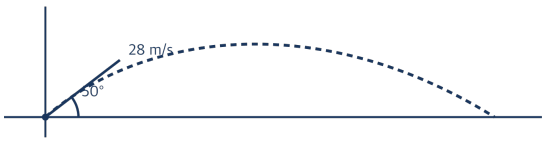
Exit ticket

Projectile Motion

Components & Basics

Name _____

EC1



A projectile is launched from level ground at 28 m/s at 50°. Find the total time of flight. Take $g = 9.8\ m/s^2$.

Show your working

KAIZEN MATHS

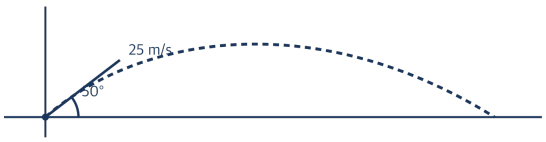
Exit ticket

Projectile Motion

Components & Basics

Name _____

EA1



A projectile is launched at 25 m/s at an angle of 50° above the horizontal. Find its horizontal and vertical components of velocity.

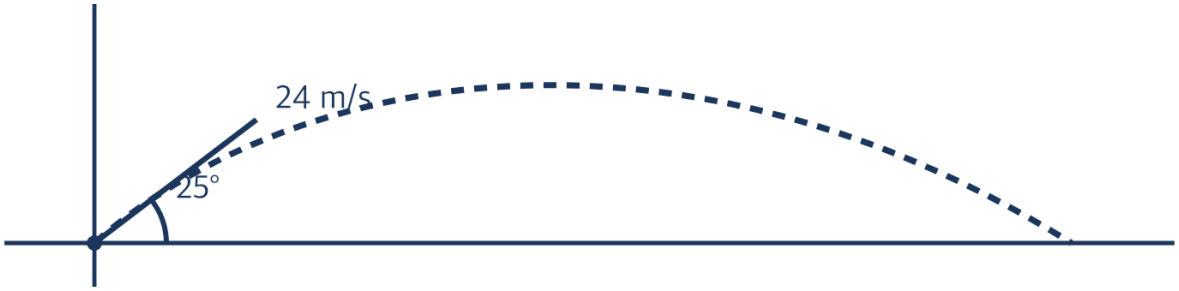
Show your working

Teacher answers & working

Projectile Motion · Components & Basics

MEC-5B63-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



A projectile is launched at 24 m/s at an angle of 25° above the horizontal. Find its horizontal and vertical components of velocity.

Answer:

$u_x = 21.75$ m/s, $u_y = 10.14$ m/s

Working:

Resolve the initial velocity horizontally and vertically.

$u_x = u \cos\theta$ and $u_y = u \sin\theta$

Substitute u = 24 and θ = 25°.

$u_x = 24 \cos 25° = 21.75$ m/s

$u_y = 24 \sin 25° = 10.14$ m/s

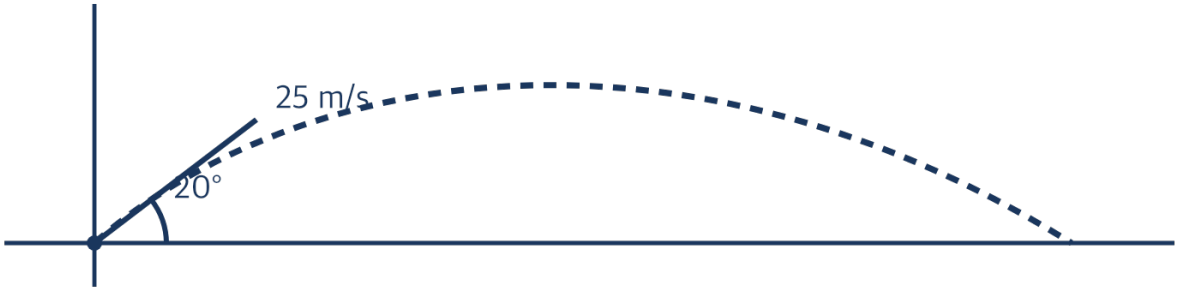
The components are u_x = 21.75 m/s and u_y = 10.14 m/s.

Teacher answers & working

Projectile Motion · Components & Basics

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SB1



A projectile is launched at 25 m/s at 20° above the horizontal. Find the time taken to reach maximum height.

Answer:

$t = 0.87$ s

Working:

Find the initial vertical component.

$u_Y = 25 \sin 20° = 8.55$ m/s

At maximum height, the vertical velocity is 0. Use $v_y = u_y - 9.8t$.

$0 = 8.55 - 9.8t$

$t = 8.55 \div 9.8 = 0.87$ s

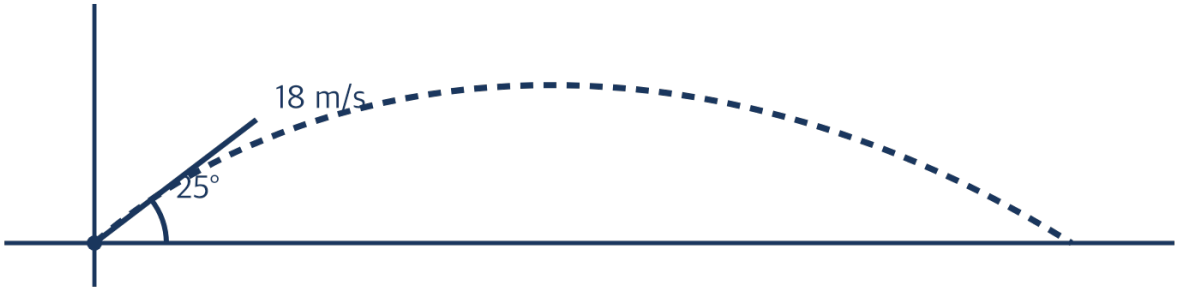
The time to maximum height is 0.87 s.

Teacher answers & working

Projectile Motion · Components & Basics

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SC1



A projectile is launched from level ground at 18 m/s at 25°. Find the total time of flight. Take g = 9.8 m/s².

Answer:

$T = 1.55$ s

Working:

Resolve the initial vertical velocity.

$u_Y = 18 \sin 25° = 7.61$ m/s

On level ground, the projectile returns to y = 0. Use $y = u_y t - 4.9t^2$.

$0 = 7.61t - 4.9t^2$

$t(7.61 - 4.9t) = 0$

$t = 0$ or $t = 7.61 \div 4.9 = 1.55$

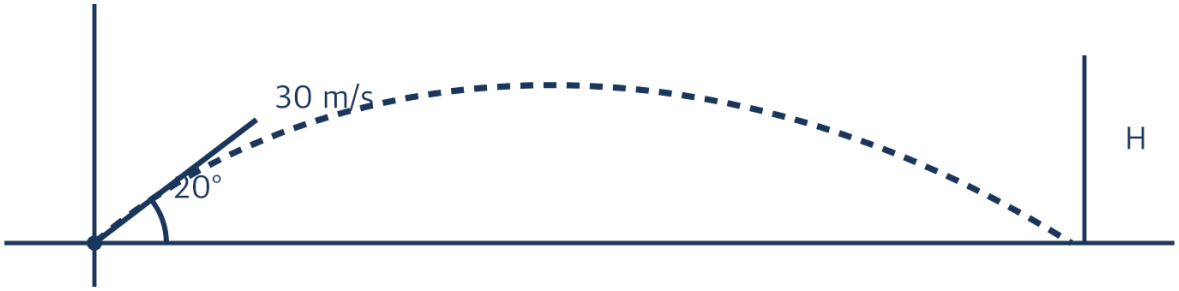
Ignoring the launch time t = 0, the total time of flight is 1.55 s.

Teacher answers & working

Projectile Motion · Components & Basics

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SD1



A projectile is launched from level ground at 30 m/s at 20°. Find the maximum height reached.

Answer:

$H = 5.37$ m

Working:

Resolve the initial vertical velocity.

$u_y = 30 \sin 20° = 10.26$ m/s

At maximum height, $v_y = 0$. Use $v_y^2 = u_y^2 + 2as$.

$0^2 = 10.26^2 + 2 \times (-9.8) \times H$

$19.6H = 105.28$

$H = 105.28 \div 19.6 = 5.37$ m

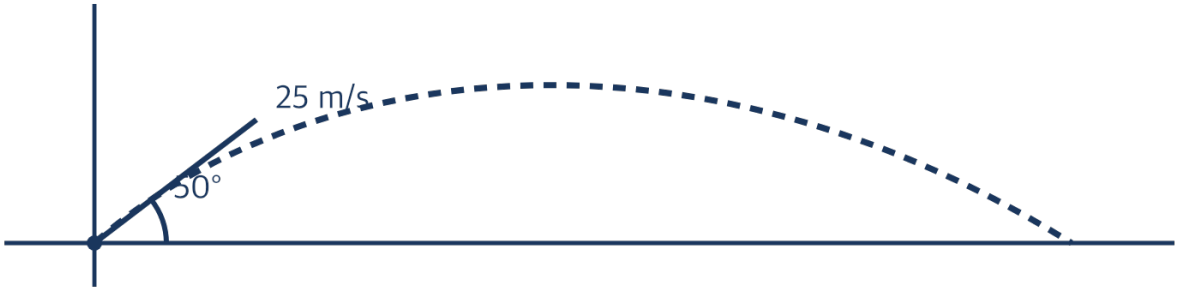
The maximum height is 5.37 m.

Teacher answers & working

Projectile Motion · Components & Basics

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EA1



A projectile is launched at 25 m/s at an angle of 50° above the horizontal. Find its horizontal and vertical components of velocity.

Answer:

$u_x = 16.07$ m/s, $u_y = 19.15$ m/s

Working:

Resolve the initial velocity horizontally and vertically.

$u_x = u \cos\theta$ and $u_y = u \sin\theta$

Substitute u = 25 and θ = 50°.

$u_x = 25 \cos 50° = 16.07$ m/s

$u_y = 25 \sin 50° = 19.15$ m/s

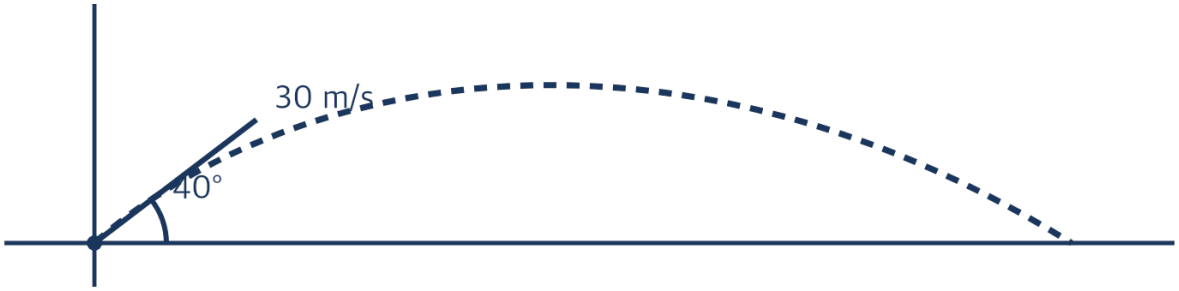
The components are u_x = 16.07 m/s and u_y = 19.15 m/s.

Teacher answers & working

Projectile Motion · Components & Basics

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EB1



A projectile is launched at 30 m/s at 40° above the horizontal. Find the time taken to reach maximum height.

Answer:

$t = 1.97$ s

Working:

Find the initial vertical component.

$u_Y = 30 \sin 40° = 19.28$ m/s

At maximum height, the vertical velocity is 0. Use $v_y = u_y - 9.8t$.

$0 = 19.28 - 9.8t$

$t = 19.28 \div 9.8 = 1.97$ s

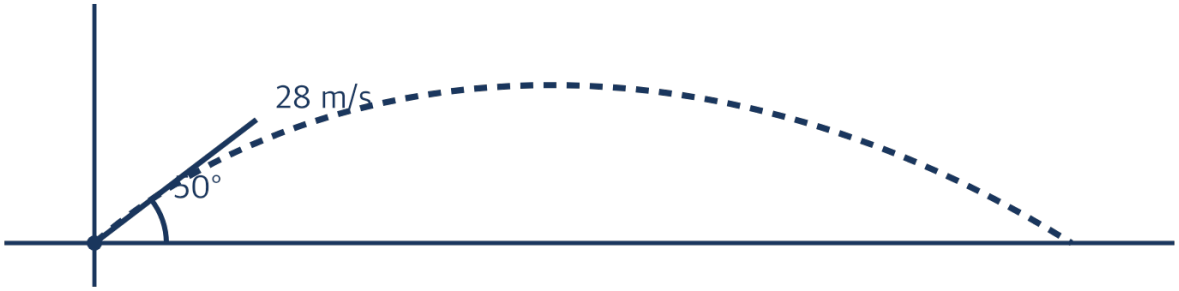
The time to maximum height is 1.97 s.

Teacher answers & working

Projectile Motion · Components & Basics

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EC1



A projectile is launched from level ground at 28 m/s at 50°. Find the total time of flight. Take g = 9.8 m/s².

Answer:

$T = 4.38$ s

Working:

Resolve the initial vertical velocity.

$u_Y = 28 \sin 50° = 21.45$ m/s

On level ground, the projectile returns to y = 0. Use $y = u_Y t - 4.9t^2$.

$0 = 21.45t - 4.9t^2$

$t(21.45 - 4.9t) = 0$

$t = 0$ or $t = 21.45 \div 4.9 = 4.38$

Ignoring the launch time t = 0, the total time of flight is 4.38 s.